



EDUCATION

University Institute of Engineering and Technology

2022 - 2027

B.E. - Mechanical Engineering | CGPA: 6.07 / 10

Gov Model Sen Sec School, Chandigarh

2022

12th | cbse | Percentage: 64.40 / 100

St Xavier Sen Sec School, Chandigarh

2020

10th | ICSE | Percentage: 69.60 / 100

INTERNSHIPS

IIT ROPAR | Education

29 Dec, 2025 - 16 Jan, 2026

Intern

Key Skills: Stability Assesement Matlab Simulation Python automation TPMS Modelling

Developed a research-oriented MATLAB application for parametric generation, visualization, and export of TPMS and other equation-driven complex geometries, with structured backend logic and emphasis on usability and reproducibility.

Implemented analytical geometry formulations within MATLAB to generate simulation compatible .stl files, enabling controlled variation of geometric parameters for design and analysis studies.

Designed interactive workflows within the application to manage geometric inputs, automate shape construction, and support systematic dataset creation.

Built modular app components to allow scalable extension of geometry types and future research functionalities.

Assisted a PhD researcher with selected aspects of mechanical characterization studies, contributing to experimental data handling and validation support.

Processed experimental datasets using Excel and plot digitization tools to extract key mechanical response parameters.

Supported numerical studies by organizing and mapping experimental and geometric data into Abaqus-compatible input formats.

Maintained structured datasets to ensure consistency across geometry generation, experiments, and simulations.

Prepared technical plots, summaries, and presentations to communicate geometric trends, modeling outcomes, and research progress within the research group.

IIT ROPAR | Education

14 Jul, 2025 - 04 Aug, 2025

Summer Intern

Key Skills: 2d mathematical modelling Abaqus

Developed and validated nanoindentation characterization workflows using finite-element models in Abaqus to reproduce indentation mechanics, extract material properties, and analyze quasi-static response.

Processed experimental data through parameter extraction and structured data mapping, converting raw measurements into simulation-ready datasets and refining modeling assumptions.

Automated Abaqus pre- and post-processing through Python scripting, enabling batch simulations, efficient data transfer, and streamlined analysis.

Built reduced-order two-dimensional analytical models to approximate three-dimensional behavior, assessed underlying assumptions, and verified accuracy against higher-fidelity simulations.

Contributed to professor-led research by preparing technical presentations, digitizing and plotting data, and synthesizing relevant literature to support and contextualize results.

Maintained detailed documentation of methodologies and findings, including modeling notes, plots, and presentation material, to ensure clarity and continuity in ongoing research.

Conducted focused analysis of Johnson–Cook constitutive modeling, examining its applicability and limitations in static and quasi-static solver formulations.

NIT Trichy | Education

07 Jun, 2025 - 07 Jul, 2025

Summer Intern

Key Skills: Abaqus Thermal simulation Johnson-Cook constitutive modeling Simulation

Developed finite-element models to support experimental metal cutting studies, with careful alignment of simulated contact conditions and experimentally observed cutting responses. Implemented thermomechanical material behavior using the Johnson–Cook constitutive framework to capture deformation, strain-rate sensitivity, and thermal effects in simulations. Evaluated numerical performance in Abaqus by examining convergence trends, mesh dependency, solver termination cases, and the effect of temperature evolution on result accuracy.

Performed mechanical testing and metallographic examination of stainless-steel struts, including UTM tensile tests, progressive surface preparation, and microstructural inspection to relate structure with mechanical behavior.

Observed robotic additive manufacturing processes involving a six-axis vertical arm and documented key process variables for use in simulation studies.

Compiled technical documentation covering background research, modeling procedures, and simulation workflows while maintaining confidentiality of experimental outcomes.

PROJECTS

Worm & Wheel Project

25 Apr, 2023 - 28 Apr, 2023

Mentor: Mr. Amandeep Singh Wadhwa | **Team Size:** 5

Key Skills:

Mechanical assembly , Basic gear and worm mechanism understanding Shaft and component alignment

Gear meshing and positioning Hands-on prototype fabrication Manual cutting and shaping

Designed and fabricated a working worm–worm wheel mechanism with a load-lifting arrangement to demonstrate speed reduction, torque multiplication, mechanical advantage and efficiency; manually assembled and aligned the shaft, worm, gear, pulley and load system.

POSITIONS OF RESPONSIBILITY

Joint Secretary

Duration: (Oct,2024 - Nov,2025)

I joined the **JUGAAD Robotics Club in 2024** and worked as a Technical Team Member for a year before taking up the role of **Joint Secretary**. In this role, my main responsibilities include **event management, member coordination, backend planning, and communication with speakers**. I also handle **official documentation, drafting letters, getting necessary approvals and signatures from professors, preparing letterheads, and assigning roles to volunteers during events**. Along with the management side, I continue to contribute to the **technical work of the club** and coordinate closely with the core team and members to maintain smooth communication and teamwork.

Duration:

PERSONAL DETAILS

Gender: Male

Marital Status: Single

Current Address: Flat no 18 b decent homes 01 sector 117 mohali punjab, Mohali, Punjab, India - 140307

Email: akshitdewan211@gmail.com

Date of Birth: 02 Nov, 2004

Known Languages: Hindi English Panjabi

Phone Number: +91-7380159085

REFERENCES

- **Dr Samir Chnadra Roy**
(Assistant Professor, IIT ROPAR)
(scroy@iitrpr.ac.in)

IMs

- Other - <https://www.linkedin.com/in/akshit-dewan-mechanical/>